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Fasteners for electrical connectors of a vehicle

Abstract

A fastener for electrical connector a vehicle is provided. The fastener includes a first component including a snap, an engagement surface and a connector. The first component is configured to removably affix to the electrical connector. The fastener also includes a second component including a protrusion, wherein the first component is disposed on top of the second component. The protrusion limits a range of motion of the snap in a direction of the second component.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8763212	12/2013	Scroggie	24/297	F16B 5/0628
10808745	12/2019	Javali	N/A	F16B 21/086
11692649	12/2022	Locatelli	248/74.1	F16L 3/233
12024098	12/2023	Chen	N/A	B60R 13/0206

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Background/Summary

INTRODUCTION

(1) The disclosure relates to fasteners, and more particularly to a fastener for an electrical connector of a vehicle.

(2) In general, vehicles include many different electrical systems. These electrical systems include, but are not limited to, infotainment systems, lighting systems, power steering systems, power braking system, driver assistance systems, various sensors, heating and air conditioning systems, and the like. Each of the electrical systems include numerous wires and connectors that are disposed throughout the vehicle. The wires and connectors are secured to a portion of the vehicle to prevent the wires and connectors from being damaged or interfering with the operations of other components of the vehicle.

SUMMARY

(3) In one exemplary embodiment, a fastener for an electrical connector of a vehicle is provided. The fastener includes a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector. The fastener also includes a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component.

(4) In addition to the one or more features described herein the second component includes one or more alignment posts.

(5) In addition to the one or more features described herein the first component includes one or more alignment apertures that are configured to receive the one or more alignment posts.

(6) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.

(7) In addition to the one or more features described herein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.

(8) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the

protrusion at the first end.

(9) In addition to the one or more features described herein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.

(10) In one exemplary embodiment, a vehicle including a fastener for an electrical connector is provided. The fastener includes a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector. The fastener also includes a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component.

(11) In addition to the one or more features described herein the second component includes one or more alignment posts.

(12) In addition to the one or more features described herein the first component includes one or more alignment apertures that are configured to receive the one or more alignment posts.

(13) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.

(14) In addition to the one or more features described herein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.

(15) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the protrusion at the first end.

(16) In addition to the one or more features described herein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.

(17) In one exemplary embodiment, a fastener for an electrical connector of a vehicle is provided. The fastener includes a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector. The fastener also includes a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component. The snap has a first end connected to the first component and a second end adjacent the engagement surface and wherein the snap has a height that increases from the first end to the second end.

(18) In addition to the one or more features described herein the snap has a width that increases from the first end to the second end.

(19) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.

(20) In addition to the one or more features described herein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.

(21) In addition to the one or more features described herein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the protrusion at the first end.

(22) In addition to the one or more features described herein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.

(23) The above features and advantages, and other features and advantages of the disclosure are

readily apparent from the following detailed description when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other features, advantages and details appear, by way of example only, in the following detailed description, the detailed description referring to the drawings in which:
- (2) FIG. 1 is a schematic diagram of a vehicle for use in conjunction with one or more embodiments of the present disclosure;
- (3) FIGS. 2A, 2B, 2C, and 2D are schematic diagrams illustrating a fastener for an electrical connector for a vehicle in accordance with an exemplary embodiment;
- (4) FIGS. 3A and 3B are schematic diagrams illustrating a fastener for an electrical connector for a vehicle in accordance with an exemplary embodiment;
- (5) FIGS. 4A and 4B are schematic diagrams illustrating a fastener for an electrical connector for a vehicle in accordance with an exemplary embodiment; and
- (6) FIGS. 5A and 5B are schematic diagrams illustrating a fastener for an electrical connector for a vehicle in accordance with an exemplary embodiment.

DETAILED DESCRIPTION

- (7) The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses.
- (8) As discussed above, vehicles include many different electrical systems that each have numerous wires and connectors. The wires and connectors are secured to a portion of the vehicle to prevent the wires and connectors from being damaged or interfering with the operations of other components of the vehicle. Currently available fasteners for electrical connectors can be damaged during service when the electrical connector is removed. In exemplary embodiments, fasteners for electrical connectors are provided which include stops that restrict the movement of a snap as it is depressed during use. By limiting the range of movement of the snap during use of the fastener, damage to the fastener due to material strain can be avoided.
- (9) Referring now to FIG. 1, a schematic diagram of a vehicle **100** for use in conjunction with one or more embodiments of the present disclosure is shown. The vehicle **100** includes an electrical system that includes one or more wires and connectors (not shown). In one embodiment, the vehicle **100** includes a door **102** that includes a plurality of electrical connectors that are secured to an inner surface of the door **102** by fasteners (not shown). Although disused primarily in connection with the use in a door, those of ordinary skill will appreciate that the electrical connectors may be secured to any interior surface of a vehicle.
- (10) Referring now to FIGS. 2A, 2B, 2C, and 2D schematic diagrams illustrating a fastener **200** for an electrical connector for a vehicle in accordance with an exemplary embodiment are shown. In general, the fastener **200** includes a first component **202** and a second component **204** and the first component **202** is configured to be disposed on top of the second component **204**. The fastener **200** is configured to receive an electrical connector (not shown) and prevent the electrical connector from moving.
- (11) As illustrated, the first component **202** includes a snap **206** that includes a first end **207** that is affixed to the first component **202** and a second end **209**. The snap **206** has a height that generally increases as the snap **206** extends from the first end **207** to the second end **209**. In exemplary embodiments, the snap **206** has a width that is wider at the first end **207** than at the second end **209**. In one embodiment, the width of the snap **206** decreases as the snap **206** extends from the first end **207** to the second end **209**. In another embodiment, the width of the snap **206** is uniform from the first end **207** to the second end **209**. In a further embodiment, the width of the snap **206** increases as

the snap **206** extends from the first end **207** to the second end **209**. In one embodiment, the length of the snap **206** is greater than 20 mm long and the snap has a thickness of 2.5 mm to avoid fracture of the snap **206** and to allow the snap **206** to bend within the permissible strain of the material used to form the snap **206**.

(12) The first component **202** also includes an engagement surface **212** adjacent to the second end **209** of snap **206** and a connector **208** disposed on the opposite side of the engagement surface **212** from the snap **206**. In one embodiment, a distance between the second end **209** of the snap **206** and the engagement surface **212** is approximately 0.5 mm. In exemplary embodiments, the engagement surface **212** has a width **213** which is less than the width **211** of the connector **208**. In exemplary embodiments, the connector **208** includes side portions **210** that extend outward from the sides of the engagement surface **212**.

(13) As illustrated, the second component **204** includes a protrusion **214**, which is also referred to herein as a rib **214**. The protrusion **214** extends upwardly from the upper surface of the second component **204**. In exemplary embodiments, the protrusion **214** has a first end **216** and a second end **218**. In one embodiment, the width of the protrusion **214** is smaller at the first end **216** than the width of the protrusion at the second end **218**. In one embodiment, the height of the protrusion **214** is shorter at the first end **216** than the height of the protrusion at the second end **218**. In exemplary embodiments, the height and shape of the protrusion **214** are determined based on the behavior of the snap **206** when it bends.

(14) The first component **202** also includes an aperture **205** disposed beneath the snap **206**. The aperture **205** is configured to receive the protrusion **214** of the second component **204** when the first component **202** is placed on top of the second component **204**. The protrusion **214** is configured to limit the movement of the second end **209** of the snap **206** towards the surface of the first component **202**, to prevent fatigue and/or failure of the snap **206** when pressed, FIG. 2D. In exemplary embodiments, the protrusion **214** limits the distance that the snap **206** can move towards the upper surface of the second component **204** to less than 2-3 mm to prevent fatigue and/or failure of the snap **206** when pressed. In an exemplary embodiment, the aperture **205** has a tapered shape that is configured to aid in the alignment of the protrusion **214** with the snap **206**. In one example, the protrusion **214** has a larger width at the second end **218** than the width at the first end **216** and the sides of the protrusion **214** are configured to come into contact with the sides of the aperture **205** to help properly align the protrusion **214** with the snap **206**.

(15) Turning now to FIGS. 3A and 3B schematic diagrams illustrating a fastener for an electrical connector for a vehicle in accordance with an exemplary embodiment are shown. In general, the fastener includes a first component **302** and a second component **304**, and the first component **302** is configured to be disposed on top of the second component **304**. The first component **302** includes an aperture **305** that is configured to receive a protrusion **314** of the second component **304** when the first component **302** is disposed on top of the second component **304**. The protrusion **314** is configured to limit the movement of a snap **306** towards the surface of the first component **302**, to prevent fatigue and/or failure of the snap **306** when pressed.

(16) The second component **304** includes one or more alignment posts **320** and apertures **322**. The first component **302** includes one or more alignment apertures **326** that are configured to receive the alignment posts **320** when the first component **302** is disposed on top of the second component **304**. In exemplary embodiments, alignment apertures **326** and the alignment posts **320** are configured to ensure that the snap **306** is properly aligned with the protrusion **314**. In exemplary embodiments, the apertures **324** are configured to overlap with the apertures **322** such that a screw, rivet, or other attachment means can be placed through both the first component **302** and the second component **304** to secure the first component **302** and the second component **304** to a surface (not shown).

(17) Turning now to FIGS. 4A and 4B schematic diagrams illustrating a fastener **400** for an electrical connector for a vehicle in accordance with an exemplary embodiment are shown. As

illustrated, the fastener **400** includes a snap **404**, a protrusion **408**, an engagement surface **406**, and a connector **402**. The snap **404** includes a first end **410** that is connected to a body **401** of the fastener **400** and a second end **412** opposite the first end **410**. The protrusion **408** is configured to limit the movement of the second end **412** of the snap **404** towards a surface **411** of the fastener **400**, to prevent fatigue and/or failure of the snap **404** when pressed. In exemplary embodiments, the engagement surface **406** has a width **413** which is less than the width **411** of the connector **402**. In exemplary embodiments, the connector **402** includes side portions **414** that extend outward from the sides of the engagement surface **406**.

(18) Turning now to FIGS. 5A and 5B schematic diagrams illustrating a fastener **500** for an electrical connector **510** for a vehicle in accordance with an exemplary embodiment are shown. As illustrated, the fastener **500** includes a first component **502** that is disposed on top of a second component **504**. The first component **502** includes a snap **506** that is configured to deform towards an upper surface **511** of the second component **504**. The second component **504** includes a protrusion **508** disposed beneath the snap **506** that limits the movement of the snap **506** towards the upper surface **511** of the second component **504**, to prevent fatigue and/or failure of the snap **506** when pressed. In one embodiment, the length of the snap **506** is approximately 20 mm long and the snap **506** has a thickness of 2.5 mm to avoid any fracture and allow the snap **506** to bend within the permissible strain of the material. In exemplary embodiments, the protrusion **508** limits the distance that the snap **506** can move towards the upper surface **511** of the second component **504** to less than 2-3 mm to prevent fatigue and/or failure of the snap **506** when pressed.

(19) In exemplary embodiments, the electrical connector **510** includes a tab **512** that is configured to depress the snap **506** as the electrical connector **510** is slid onto the fastener **500**. In one embodiment, the surface **518** of the first component **502** includes one or more rails that are configured to engage with a lower portion **514** as the connector **510** is slid into place across the snap **506**. Once the tab **512** slides past an end **516** of the snap **506**. The snap **506** moves in an upward direction, (i.e., away from the second component **504**), and prevents the tab **512** from traversing backward across the snap **506**. For extraction of the connector **510**, the snap **506** needs to be pressed downward until it contacts the ribs **508**.

(20) The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the aspect is included in at least one aspect described herein, and may or may not be present in other aspects. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various aspects.

(21) When an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

(22) Unless specified to the contrary herein, all test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

(23) Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

(24) While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the

present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof.

Claims

1. A fastener for an electrical connector of a vehicle comprising: a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector; a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component.
2. The fastener of claim 1, wherein the second component includes one or more alignment posts.
3. The fastener of claim 2, wherein the first component includes one or more alignment apertures that are configured to receive the one or more alignment posts.
4. The fastener of claim 1, wherein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.
5. The fastener of claim 1, wherein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.
6. The fastener of claim 5, wherein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the protrusion at the first end.
7. The fastener of claim 6, wherein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.
8. A vehicle comprising: a fastener for electrical connector, the fastener comprising: a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector; a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component.
9. The vehicle of claim 8, wherein the second component includes one or more alignment posts.
10. The vehicle of claim 9, wherein the first component includes one or more alignment apertures that are configured to receive the one or more alignment posts.
11. The vehicle of claim 8, wherein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.
12. The vehicle of claim 8, wherein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.
13. The vehicle of claim 12, wherein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the protrusion at the first end.
14. The vehicle of claim 13, wherein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.
15. A fastener for an electrical connector of a vehicle comprising: a first component including a snap, an engagement surface and a connector, wherein the first component is configured to removably affix to the electrical connector; a second component including a protrusion, wherein the first component is disposed on top of the second component and wherein the protrusion limits a range of motion of the snap in a direction of the second component, wherein the snap has a first end connected to the first component and a second end adjacent the engagement surface and wherein the snap has a height that increases from the first end to the second end.

16. The fastener of claim 15, wherein the snap has a width that increases from the first end to the second end.

17. The fastener of claim 15, wherein the protrusion includes a first end and a second end and wherein a height of the protrusion at the second end is greater than a height of the protrusion at the first end.

18. The fastener of claim 15, wherein the first component includes an aperture disposed beneath a portion of the snap and wherein at least a portion of the protrusion is configured to extend through the aperture.

19. The fastener of claim 18, wherein the protrusion includes a first end and a second end and wherein a width of the protrusion at the second end is greater than a width of the protrusion at the first end.

20. The fastener of claim 19, wherein the aperture has a tapered shape and wherein the first end of the protrusion and the second end of the protrusion are configured to contact sides of the aperture to aid in alignment of the snap with the protrusion.
